

Mock Test — Exploring the Investigative World of Science

A mixed paper covering the whole chapter. Total: 20 marks. Section A is interactive.

Objective (1 mark each)

Q1 • MCQ**[1]**

Which is a scientific (testable) question?

- (a) Is science fun? (b) Does adding more yeast make dough rise faster?
(c) Which colour is prettiest? (d) Should we cook daily?

Q2 • MCQ**[1]**

"The water is at 50°C" is a —

- (a) qualitative observation (b) quantitative observation (c) inference (d) question

Q3 • MCQ**[1]**

The variable a scientist deliberately changes is the —

- (a) measured variable (b) controlled variable (c) changed (independent) variable
(d) hidden variable

Q4 • TRUE / FALSE**[1]**

A fair test changes only one variable at a time.

- (a) True (b) False

Q5 • ASSERTION-REASON**[1]**

A: "The plant died because it had no light" is an inference.

R: An inference explains observations.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Q6 · FILL IN THE BLANK

[1]

The thing you measure as the result is the _____ variable.

Q7 · FILL IN THE BLANK

[1]

Doing experiments step by step is called _____ investigation.

Q8 · MCQ

[1]

Why repeat an experiment several times?

- (a) to check the result is reliable (b) to change the result (c) it is never needed
(d) to use more material

Short answer (2 marks each)

Q9

[2]

Distinguish between an observation and an inference with one example each.

Q10

[2]

Name the changed and measured variables for: "Does more fertiliser make plants taller?"

Q11

[2]

Give two good habits of a careful scientific investigator.

Long / application (3 marks each)

Q12

[3]

Design a fair experiment to test whether warm water dissolves sugar faster than cold water. State changed, measured and two controlled variables.

Q13

[3]

Read: "I left dough out for 5 hours; it doubled in size and smelled sour, probably because tiny organisms acted on it." Identify two observations and one inference, and suggest one new question it raises.

[Check yourself](#) → Answer key